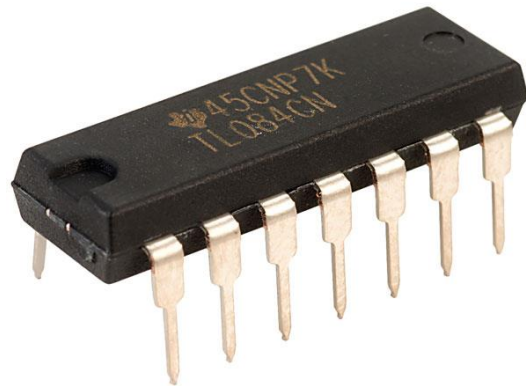


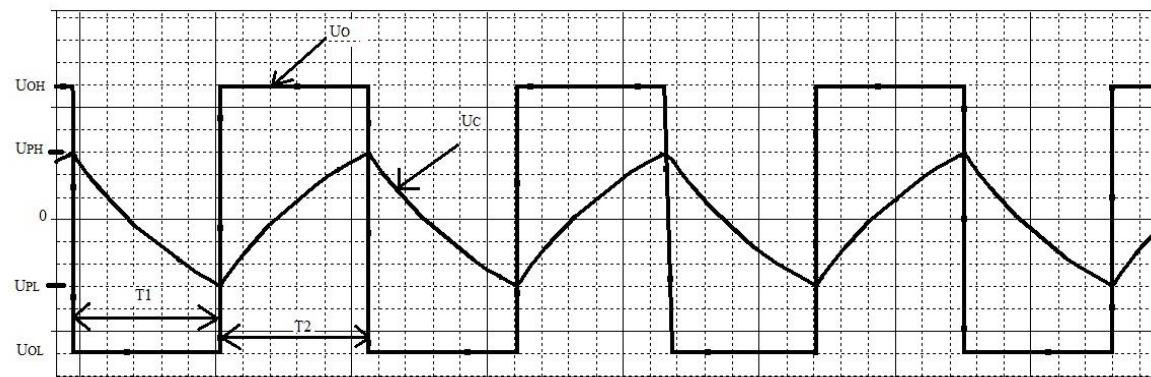
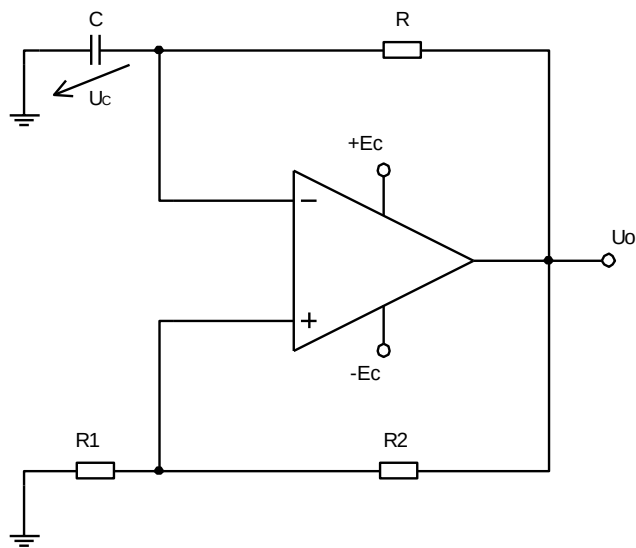
Bazele Circuitelor Electronice

Aplicații cu Amplificatoare Operaționale

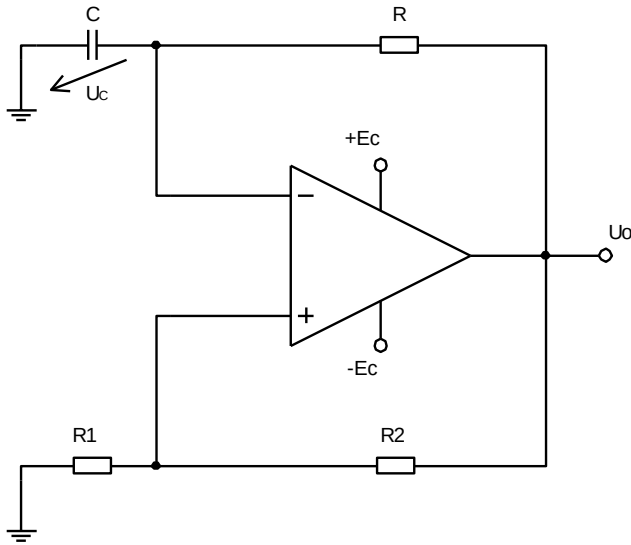


Generatoare de semnal cu AO

Astabil cu AO



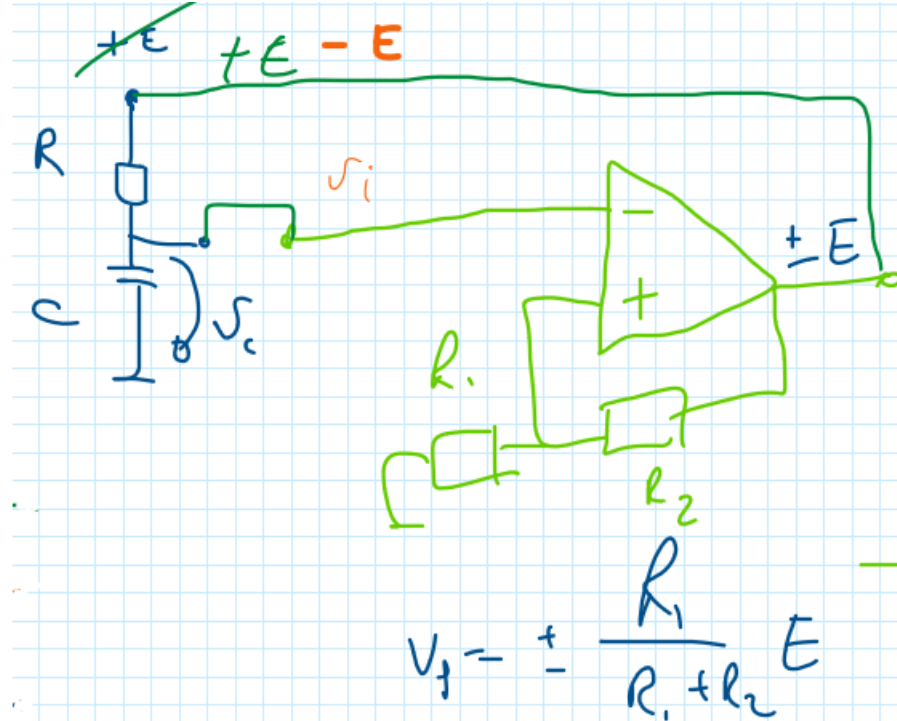
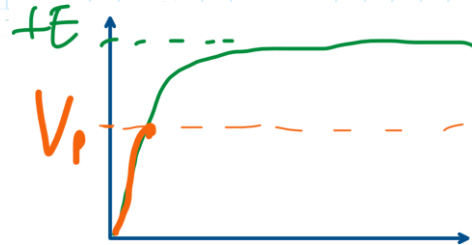
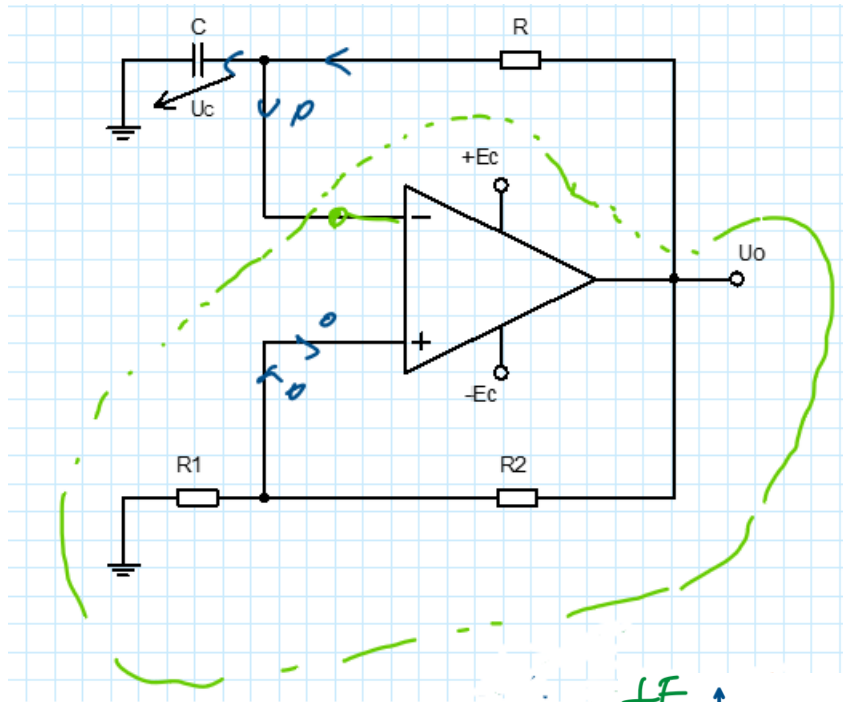
Astabil cu AO



Cir. basculante → monostabile
→ bistabil
→ astabil

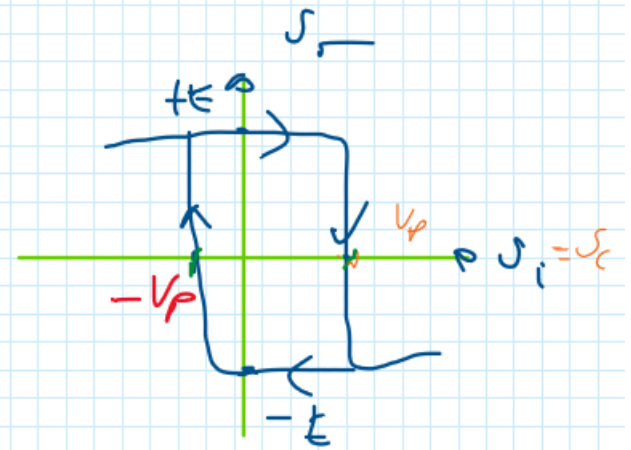


Astabil cu AO

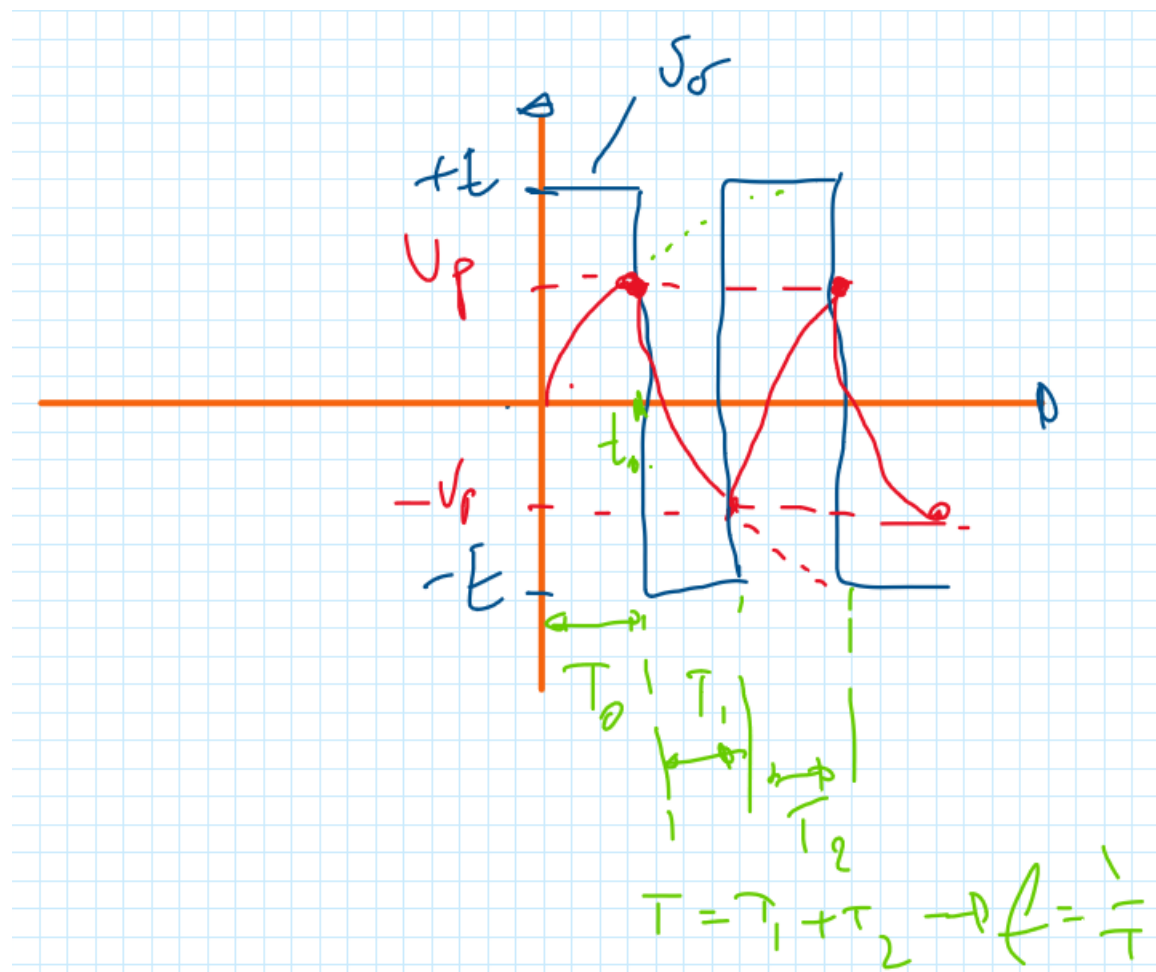


$$V_f = \pm \frac{R_1}{R_1 + R_2} E$$

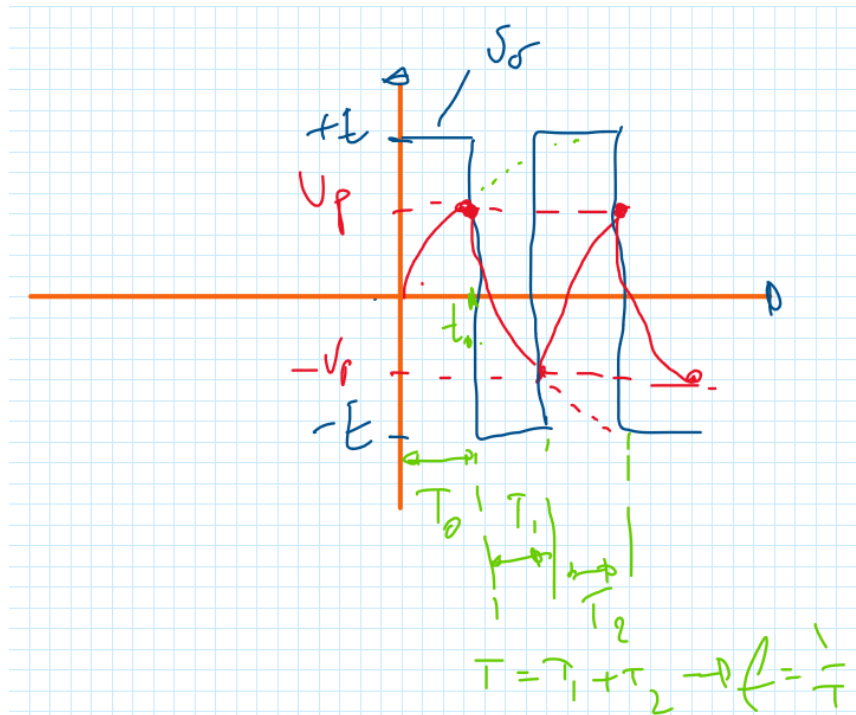
Comp. cu R. L. W.



Astabil cu AO



Astabil cu AO



$$T_0: V_c(t) = V_c(\infty) - [V_c(\infty) - V_c(0)] \times e^{-\frac{t}{\tau}}, \quad \tau = RC$$

$$V_c(T_0) = E - (E - 0) \times e^{-\frac{T_0}{\tau}}$$

$$V_c(T_0) = V_c(t_0) = V_p = \frac{R_1}{R_1 + R_2} E$$

$$\frac{R_1}{R_1 + R_2} E = E - E \times e^{-\frac{T_0}{\tau}} \quad \left| \quad \frac{1}{E} \right.$$

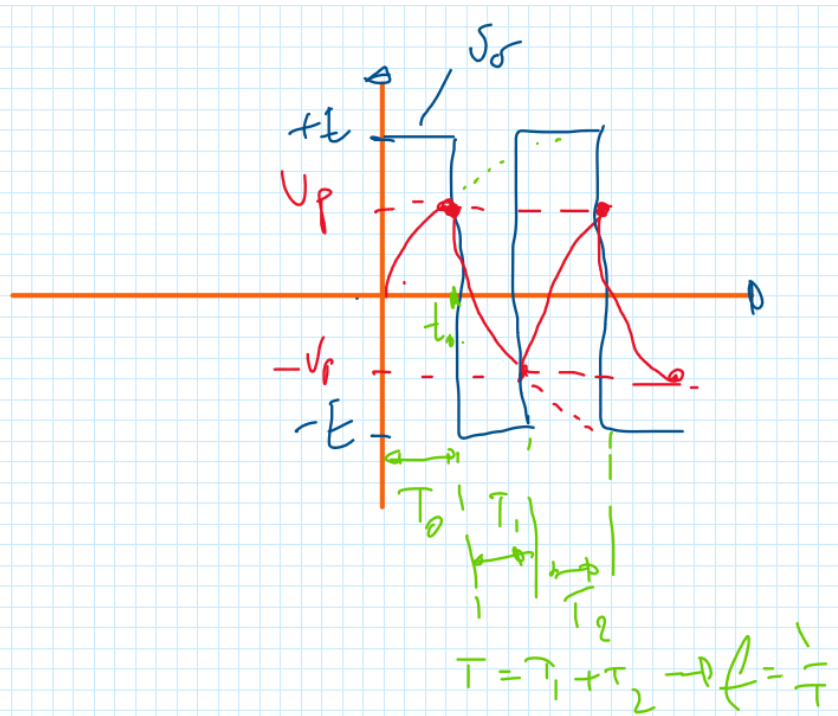
$$\frac{R_1}{R_1 + R_2} = 1 - e^{-\frac{T_0}{\tau}}$$

$$e^{-\frac{T_0}{\tau}} = 1 - \frac{R_1}{R_1 + R_2}$$

$$e^{-\frac{T_0}{\tau}} = \frac{R_2}{R_1 + R_2} \quad \left| \quad \ln \right.$$

$$T_0 = -RC \ln \frac{R_2}{R_1 + R_2}$$

Astabil cu AO



$$T_1: v_c(t) = v_c(\infty) - [v_c(\infty) - v_c(0)] e^{-\frac{t}{\tau}}$$

$$v_c(t_1) = -V_p$$

$$v_c(\infty) = -E, \quad v_c(0) = V_p$$

$$-V_p = -E - (-E - V_p) e^{-\frac{T_1}{\tau_c}}$$

$$-\frac{R_1}{R_1 + R_2} E = -E - \left(-E - \frac{R_1}{R_1 + R_2} E\right) e^{-\frac{T_1}{R_c C}} \quad \left| \frac{1}{E}\right.$$

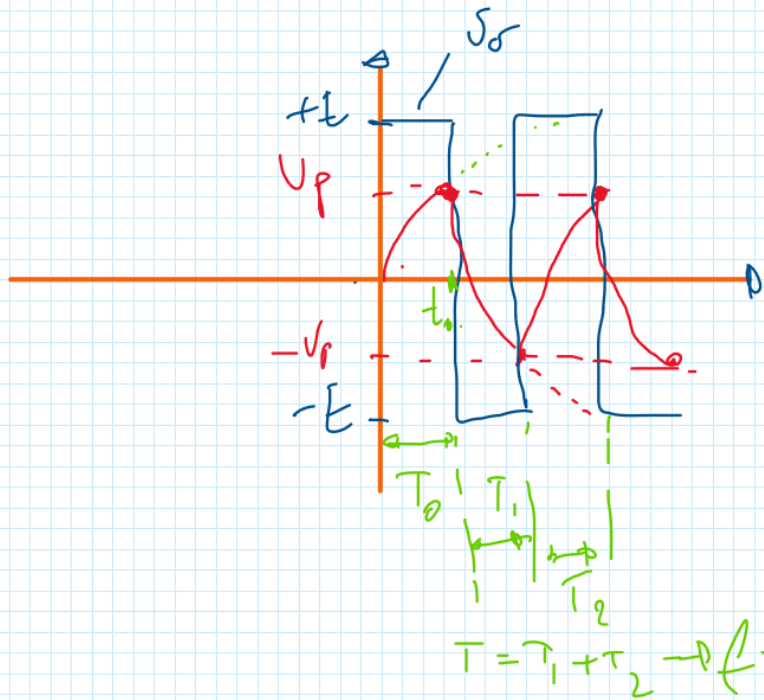
$$-\frac{R_1}{R_1 + R_2} = -1 + \left(1 + \frac{R_1}{R_1 + R_2}\right) e^{-\frac{T_1}{R_c C}}$$

$$1 - \frac{R_1}{R_1 + R_2} = \left(1 + \frac{R_1}{R_1 + R_2}\right) e^{-\frac{T_1}{R_c C}}$$

$$e^{-\frac{T_1}{R_c C}} = \frac{1 - \frac{R_1}{R_1 + R_2}}{1 + \frac{R_1}{R_1 + R_2}}$$

$$T_1 = -R_c C \ln \frac{1 - \frac{R_1}{R_1 + R_2}}{1 + \frac{R_1}{R_1 + R_2}} = -f_c C \ln \frac{R_2}{2R_1 + R_2}$$

Astabil cu AO



T_2 :

$$T_2 = T_1$$

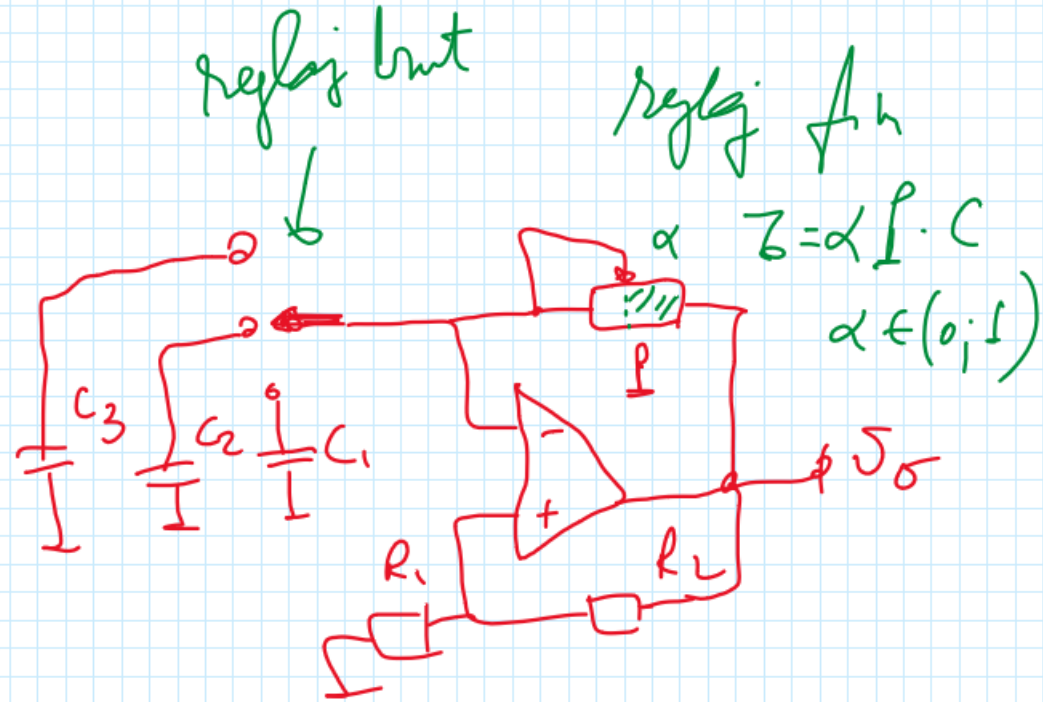
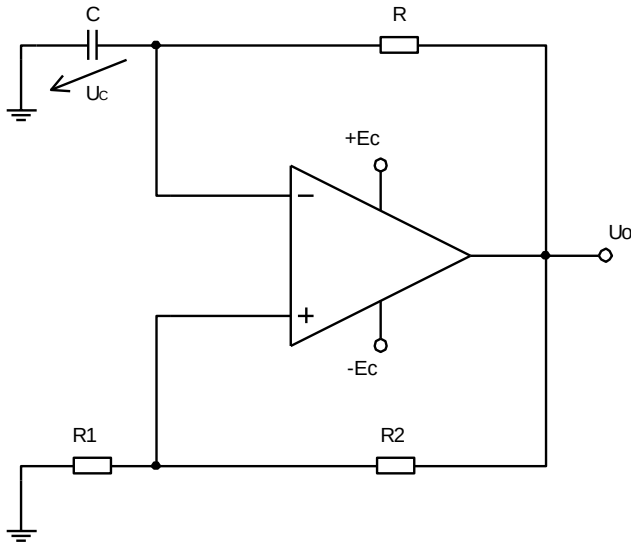
$$T = 2T_1 = -2RC \ln \frac{1 - \frac{R_1}{R_1 + R_2}}{1 + \frac{R_1}{R_1 + R_2}}$$

$$f = \frac{1}{T}$$

$$f = \frac{1}{-2RC \ln \frac{1 - \frac{R_1}{R_1 + R_2}}{1 + \frac{R_1}{R_1 + R_2}}}$$



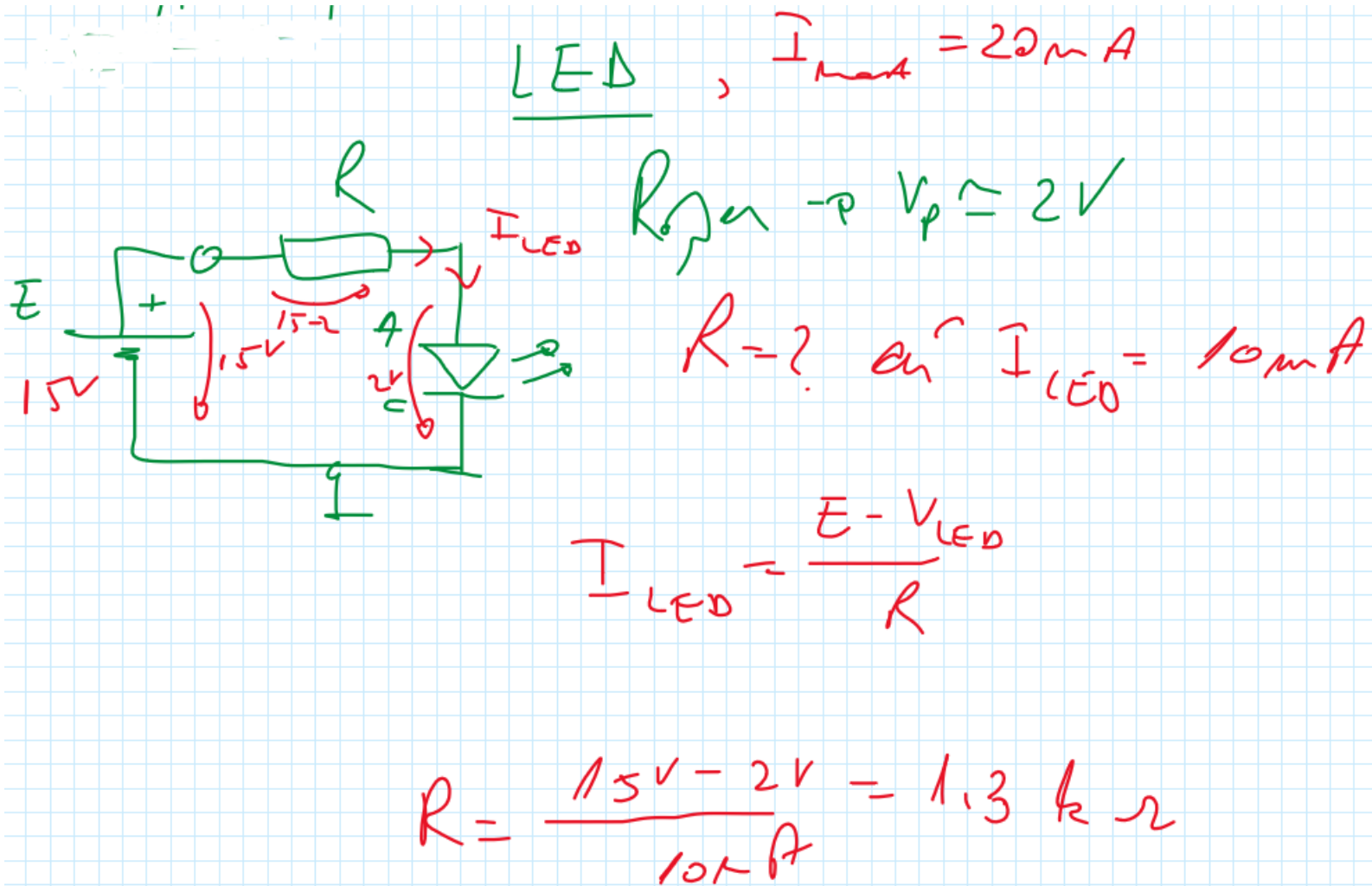
Astabil cu AO



$$C_2 = 10 C_1$$

$$C_3 = 10 C_2 - 100 C_1$$

Astabil cu AO



Astabil cu AO

LED, $I_{max} = 20mA$

$R_{LED} \rightarrow V_p \approx 2V$

$R = ?$ an $I_{LED} = 10mA$

$$I_{LED} = \frac{E - V_{LED}}{R}$$

$$R = \frac{15V - 2V}{10mA} = 1.3k\Omega$$

